$$m^2-4m-1 = 0$$

$$(m-2)^2-4-1 = 0$$

$$(m-2)^2 = 5$$

$$m-2 = \pm\sqrt{5}$$

$$m = 2\pm\sqrt{5}$$

$$C.F = Ae^{(2+\sqrt{5})x}+Be^{(2-\sqrt{5})x}$$

$$PI = \frac{1}{\phi(D)}f(x)$$

$$= \frac{1}{D^2-4D-1}e^{-3x}$$

$$= \frac{1}{(-3)^2-4(-3)-1}e^{-3x}$$

(replace D *by* −3)

$$= \frac{1}{9+12-1}e^{-3x}$$

$$= \frac{e^{-3x}}{20}$$

Hence the general solution is

$$y = C.F+P.I$$

$$\Rightarrow \quad y = Ae^{(2+\sqrt{5})x}+Be^{(2-\sqrt{5})x}+\frac{e^{-3x}}{20}$$

Example 4.30

Solve: $(D^2-2D+1)y = e^{2x}+e^x$

Solution:

$$(D^2-2D+1)y = e^{2x}+e^x$$

The auxiliary equation is

$$m^2-2m+1 = 0$$

$$\Rightarrow (m-1)(m-1) = 0$$

$$m = 1, 1$$

$$C.F = (Ax+B)e^x$$

$$PI = \frac{1}{\phi(D)}f(x) = \frac{1}{D^2-2D+1}(e^{2x}+e^x)$$

Now,

$$P.I_1 = \frac{1}{D^2-2D+1}e^{2x}$$

$$= \frac{1}{4-4+1}e^{2x}$$

(replace D *by* 2)

$$= e^{2x}$$

and

$$P.I_2 = \frac{1}{D^2-2D+1}e^x$$

$$= \frac{1}{(D-1)^2}e^x$$

Replace D by 1. $(D-1)^2 = 0$ when D = 1

$$\therefore P.I_2 = x.\frac{1}{2(D-1)}e^x$$

Replace D by 1. $(D-1) = 0$ when D = 1

$$\therefore P.I_2 = x^2\frac{1}{2}e^x$$

The general solution is

$$y = C.F+P.I_1+P.I_2$$

$$y = (Ax+B)e^x+e^{2x}+\frac{x^2}{2}e^x$$

Example 4.31

Solve: $(3D^2+D-14)y = 4-13e^{\frac{-7}{3}x}$

Solution:

$$(3D^2+D-14)y = 4-13e^{\frac{-7}{3}x}$$

The auxiliary equation is

$$3m^2+m-14 = 0$$

$$(3m+7)(m-2) = 0$$

$$m = \frac{-7}{3}, 2$$

$$C.F = Ae^{\frac{-7}{3}x}+Be^{2x}$$

$$\Rightarrow PI = \frac{1}{\phi(D)}f(x) = \frac{1}{3D^2+D-14}\left(4-13e^{\frac{-7}{3}x}\right)$$

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